

LIEKSA LAMPELA, Finland

WMO: 027960

Lat: 63.321N

Lon: 30.046E

Elev: 322

StdP: 14.53

Time Zone: 2.00 (EUE)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-21.1	-15.6	-26.8	1.2	-21.0	-20.8	1.8	-15.4	16.7	26.1	15.4	26.1	0.7	100	0.628

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.5	80.0	64.5	76.3	62.8	72.9	61.6	68.0	75.7	65.6	72.8	63.5	70.4	6.3	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.2	94.2	71.8	62.8	86.4	69.3	60.7	80.1	66.4	32.5	75.6	30.6	72.9	29.0	70.5	76.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 15.5	13.7	12.1	DB	-24.8	83.6	7.1	6.1	-30.0	88.0	-34.1	91.5	-38.1	95.0	-43.3	99.4	(4)
(5)			WB	-24.4	69.8	7.0	3.5	-29.5	72.3	-33.6	74.3	-37.5	76.3	-42.7	78.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	37.3	12.4	16.5	23.1	34.7	47.8	55.4	61.1	57.7	49.2	37.5	29.7	21.2
	DBStd	18.70	14.64	13.60	10.13	7.25	8.18	6.92	6.17	6.02	6.12	6.54	8.54	12.81
	HDD50	5555	1165	939	833	459	140	26	3	9	89	390	610	892
	HDD65	10160	1631	1359	1298	908	536	296	150	236	475	854	1060	1357
	CDD50	923	0	0	0	1	71	189	348	249	64	1	0	0
	CDD65	50	0	0	0	0	2	8	30	10	0	0	0	0
	CDH74	591	0	0	0	0	55	113	316	107	0	0	0	0
	CDH80	101	0	0	0	0	3	15	65	18	0	0	0	0
Wind	WSAvg	5.4	4.5	5.2	5.9	5.2	5.8	5.8	4.9	4.9	5.2	6.1	5.9	5.3
Precipitation	PrecAvg	25.9	1.7	1.4	1.3	1.3	2.0	2.7	3.3	3.0	2.6	2.4	2.1	2.0
	PrecMax	34.4	3.0	2.4	2.5	2.6	3.9	5.2	6.3	6.5	5.3	5.4	4.6	3.6
	PrecMin	21.4	0.6	0.3	0.5	0.2	0.6	1.1	0.8	0.9	0.6	0.6	0.5	0.4
	PrecStd	3.2	0.6	0.6	0.5	0.5	0.9	1.0	1.5	1.3	1.2	1.1	0.9	0.8
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	36.8	39.2	46.4	62.1	79.6	82.0	85.7	82.2	68.2	54.3	47.7	40.6
		MCWB	35.4	36.4	38.4	49.7	58.4	63.9	70.2	67.3	60.3	52.2	46.0	39.9
	2%	DB	34.6	36.3	40.9	55.7	74.5	77.4	81.0	76.7	64.4	51.1	44.1	36.9
		MCWB	33.6	34.4	34.8	44.8	57.1	63.0	66.6	63.8	57.3	48.7	42.7	35.6
	5%	DB	33.0	34.3	37.6	50.9	69.2	72.8	77.2	72.4	61.3	49.1	41.9	35.2
		MCWB	32.3	33.3	33.9	41.7	55.6	60.1	64.7	62.6	56.4	46.9	41.0	34.1
	10%	DB	31.7	33.0	35.6	46.8	63.8	69.0	73.7	68.9	59.0	47.0	39.2	33.8
		MCWB	31.1	32.1	32.9	39.8	51.7	58.3	63.3	61.0	55.2	45.1	38.2	33.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	36.0	36.7	39.4	50.4	61.5	68.2	72.0	69.8	61.9	52.8	46.3	40.0
		MCDB	36.7	38.5	44.7	59.5	74.3	75.7	82.6	79.9	66.7	54.1	47.3	40.4
	2%	WB	34.0	34.8	36.3	46.9	59.1	65.1	69.0	65.9	59.4	49.8	43.0	35.8
		MCDB	34.6	35.8	39.4	54.3	70.4	73.6	77.8	71.9	62.4	50.5	43.9	36.5
	5%	WB	32.5	33.4	34.7	42.7	56.3	62.1	67.0	63.8	57.3	47.2	41.1	34.3
		MCDB	33.0	34.2	37.0	50.0	67.1	70.2	74.6	70.5	60.5	48.5	41.9	34.8
	10%	WB	30.9	32.3	33.4	40.3	53.2	59.6	64.9	62.1	55.5	45.2	38.5	33.2
		MCDB	31.6	33.2	35.3	45.3	62.2	67.0	71.8	68.1	58.5	46.9	39.2	33.8
Mean Daily Temperature Range	5% DB	MDBR	11.9	12.8	16.9	17.5	19.4	17.7	17.5	16.4	13.6	9.3	7.1	9.0
		MCDBR	6.3	8.3	15.0	25.2	29.4	24.4	23.9	23.1	18.1	11.9	8.2	6.1
		MCWBR	6.2	7.7	11.2	15.2	14.4	11.8	11.2	12.2	10.2	7.8	6.1	
	5% WB	MCDBR	6.4	7.7	12.4	22.7	26.1	21.7	21.3	18.9	15.7	11.1	8.1	6.1
		MCWBR	6.3	7.4	9.8	14.0	13.7	11.3	11.0	10.8	11.1	10.1	7.8	6.2
Clear-Sky Solar Irradiance	taub	0.226	0.284	0.312	0.349	0.351	0.347	0.373	0.354	0.332	0.310	0.241	0.211	
	taud	1.984	2.100	2.157	2.280	2.444	2.484	2.426	2.465	2.513	2.390	2.075	1.966	
	Ebn at Noon	119	200	245	263	274	278	265	261	245	197	113	58	
	Edh at Noon	10	21	30	33	31	30	31	28	22	16	9	5	
All-Sky Solar Radiation	RadAvg	57	234	696	1164	1503	1659	1567	1188	655	245	59	24	
	RadStd	7	52	83	125	172	107	161	130	73	39	11	3	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (25 neighbors)		+0.97	N/A	N/A	N/A	N/A	N/A	-335	N/A	N/A	N/A

Nomenclature: See separate page